

Artificial Intelligence (CT653 / CT710)

Concise PYQ

July 22, 2026

Notes: TU IOE PYQ 2069–2081. Regular = **bold**, Back = normal; BEI/CT710 = **mono**. Marks in (); unique values only. Keys: (I)=Importance/Need (+)=Advantage (-)=Disadvantage +Fig=Figure +Eg=Example Vs=Compare/Differentiate.

1. Introduction

- Define AI; AI Vs Natural (human) intelligence; can AI replace humans (7)(6)(1+3+4)
- Turing test (I); measure machine intelligence; passed $\Rightarrow$ intelligent? (7)(2+5)
- Intelligent / rational agent; types +Eg; PEAS (Medical Diagnosis) (1+6)(2+6)(2+2+3)
- History of AI +chronological (2+5)(2+6)
- Applications of AI / fields in daily life (7)(6)
- Knowledge Vs learning; acting humanly (2+5+1)
 - Learning essential for agents +role (4+4)

2. Problem Solving

- Well-defined problem / problem space +state-space framework (3+4)
- Problem-solving steps — water jug 3/5/9 gal, +production rules +search tree (2+1+2+3)
- Farmer goat/wolf/cabbage — formulate + solve +tree (8)
- CSP crypto-arithmetic (7)(1+6)(2+6)(3+5)(5+3)
 - LOGIC+LOGIC=PROLOG; RIGHT+RIGHT=WRONG;
 - WRONG+WRONG=RIGHT
 - ONE+ONE+TWO=FOUR; SEND+MORE=MONEY;
 - SWIM+WEAR=RELAX
 - BASE+BALL=GAMES; TEN+TEN+FORTY=SIXTY;
 - LOVE+LOVE=HATE; EAT+THAT=APPLE
- Production system +Eg; why searching needed (2+6)

3. Search Techniques

- Necessity of searching +justify (2+7)
- BFS Vs DFS +performance; depth-limit search (+) (9)(4+4)(4+5)(1+7)
- Hill climbing +problems +solutions; simulated annealing; local maxima (9)(6+2)(3+5)(4+4)
- A* path+heuristic +Eg; greedy drawbacks Vs A* (8)(2+7)(2+6)
- Minimax +alpha-beta pruning +Fig(tree) (2+6)(2+4+3)

4. Knowledge Rep., Inference & Reasoning

- CNF conversion steps +Eg (10)(8)(2+6)
- FOPL need + rules of inference; horn clause; skolemization +Eg (2+3+3)(1+7)(2)
- Resolution refutation proofs (7)(8)(10)
 - American/weapons-criminal; oversmart-Ram; block-world supports
 - John/Sita-peanuts/mushroom; pompeian-Caesar; Charlie-horse/mammal

- Mary/John-study; Ram-peanuts; Sneha-happy
- Forward Vs backward chaining +Eg; rule-based reasoning (4+4)(7)(2+5)
- Bayes / causal / belief network reasoning +Eg (2+5)(2+6)(7)(3+5)(8)(5+3)
 - meningitis/stiff-neck; flu/measles rash; county/cigar; prior Vs posterior

5. Structured Knowledge Representation

- KR models / approaches / issues; how KR evaluated (7)(2+5)(3+5)(2+6)
- Semantic net +Eg; convert sentences $\rightarrow$ net (8)(7)(2+6)
- Frames +Eg; frame Vs semantic net; convert net $\rightarrow$ frame (7)(4+4)(3+5)
- Conceptual dependency + primitives (2+5)

6. Machine Learning

- Methods of learning; learning by analogy (3+5)(1+7)
- Inductive learning / ID3 +Eg (2+6)
- Genetic algorithm — steps / operators +Fig +Eg; mutation (I); applications (10)(2+8)(8)(2+6)(2+6+2)(3+5)
- Fuzzy logic / learning; Fuzzy (Mamdani) inference +Eg; membership (4)(2+6)(3+7)(8)
- Boltzmann machine +algo (1+7)(2+8)

7. Applications of AI

Neural Networks

- Backprop algo; perceptron; math model + train NN; ANN non-linearity (4+4)(2+4+4)(3+5)(2+2+4)(1+7)
- Supervised Vs unsupervised; ANN Vs human brain (6)(3+5)
- Perceptron as OR/XOR gate; McCulloch–Pitts AND (not XOR) (2+7)(1+7)(8)(2+6)
- Hebbian network (AND); Hopfield +steps +Eg; SOM +steps +Eg; Hopfield Vs Kohonen (3+7)(8)(2+5)

Expert System

- Development steps / stages +Fig block; create scenario (8)(4+4)(2+6)
- Architecture; characteristics; inference (2+4)(2+5)(3+5)
- Declarative Vs procedural knowledge + architecture (2+6)(3+6)
- (+)(-) of ES; knowledge acquisition +Eg +architecture (8)(2+6)(2+5)

NLP & Machine Vision

- NLP steps +Fig +Eg; NLU/NLG (6)(8)(2+4)(3+5)(2+7)
- Machine translation +challenges; NLP issues (1+7)(2+6)
- Syntactic / semantic / pragmatic / morphological analysis; problems (2+2+2+2)(3+5)(2+5)
- Machine / computer vision (I) +Eg (3+5)(2+5)